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Effect of V content on microstructure and mechanical properties of the $CoCrFeMnNiV_x$ high entropy alloys

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ABSTRACT

Crystal structure, microstructure, microhardness and compression properties of $CoCrFeMnNiV_x$ (x = 0, 0.25, 0.5, 0.75, 1) high entropy alloys were examined. The alloys were produced by vacuum arc melting and studied in as-solidified and homogenized (annealing at 1000 °C for 24 h) conditions. The CoCrFeMnNi alloy was a single-phase fcc solid solution in both conditions. The $CoCrFeMnNiV_{0.25}$ alloy had a single-phase fcc structure in as-solidified condition, but ~2 vol.% fine particles of a sigma phase precipitated after annealing. The alloys with x = 0.5, 0.75 and 1.0 contained the sigma phase already in as-solidified condition. The sigma-phase volume fraction increased with an increase in the V content, and in CoCrFeMnNiV the sigma phase became the matrix phase. After homogenization treatment, the volume fraction of the sigma phase increased in all three alloys by ~8% due to additional precipitation of fine particles inside the fcc phase. Phase composition and microstructure of the alloys was analyzed employing criteria for solid solution/intermetallic phase formation. The effect of alloys' chemical composition on the volume fraction of constitutive phases was discussed. A modified valence electron concentration (VEC) criterion, which takes into account localized lattice distortions around V atoms, was suggested to correctly predict sigma phase formation in the $CoCrFeNiMnV_x$ alloys. It was demonstrated that the volume fraction of sigma phase was proportional to the cumulative Cr and V concentration. Mechanical properties of the alloys were greatly affected by the sigma phase. The CoCrFeMnNi and $CoCrFeMnNiV_{0.25}$ alloys were soft and ductile, but an increase in the sigma-phase volume fraction resulted in continuous strengthening and loss of ductility.

1. Introduction

High entropy alloys (HEAs) are the new class of materials defined as alloys consisting of 5 or more principal elements with nearly equimolar fractions [1–3]. High entropy of random mixing of the principle alloying elements is thought to prevent formation of ordered solid solutions and intermetallic phases [1]. Thus, the alloys would consist mainly from simple solid solutions. However, there are experimental evidences that complex multiphase structures with a number of phases including ordered solid solutions and/or intermetallic phases may exist in many HEAs [4–6]. This indicates that the high entropy of mixing is not sufficient to prevent formation of intermetallic phases in favor of solid solution phases in HEAs [7].

The majority of reported HEAs compositions are based on the transition metals, namely Co, Cr, Fe and Ni, with addition of elements like Al, Cu, Mn, V, Ti, and Mo [1–13]. Among these alloys, a truly single phase solid solution structure based on an fcc lattice is observed only in the CoCrFeNi and CoCrFeMnNi alloys [7,13–17]. Additions of other elements to CoCrFeNi, like Al, V, Mo, and Ti cause formation of ordered or intermetallic phases [7–13]. The reasons for destabilization of a solid solution phase by alloying with different elements are not completely clear now. A number of attempts have been made [18–21] to develop criteria based on Hume-Rothery rules [22] and/or thermodynamic parameters and predict formation of solid solution or intermetallic phases in HEAs. However, strict methodology has not been developed yet. It might seem doubtful that simple approaches like Hume-Rothery rules

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would work well for such complex compositional alloys like HEAs with a variety of possible interactions between atoms of constitutive elements.

New insights into the effect of chemical composition on phase development in HEAs can be obtained by studying the effect of alloying of a "base" single phase solid solution alloy, like CoCrFeNi or CoCrFeMnNi, with various concentrations of elements causing formation additional phases. Many reported HEAs have equiatomic or near equiatomic compositions, while varying concentrations of individual elements can significantly affect both microstructure and properties of HEAs [2]. For example, the effect of Al addition on the microstructure of $CoCrFeNiAl_x$ and $CoCrFeMnNiAl_x$ alloys was systematically studied, and transition was demonstrated from single-phase fcc to single-phase bcc structures through a mixture of both phases with an increase in the Al content [10,11]. Formation of the bcc phase resulted in significant strengthening of the alloys. However, similar systematic studies are limited to Al, and are not available for many other possible elements. It should be noted that such studies can be useful not only for understanding of solid solution or ordered/intermetallic phase formation in HEAs, but also for establishing compositions with promising properties. For example, a second phase soluble at high temperatures can make a solid-solution based HEA heat treatable.

One of the possible alloying elements causing intermetallic phase formation in the CoCrFeMnNi and CoCrFeNi alloys is V. It was recently reported that V forms a sigma phase when added to these alloys at equimolar concentrations [13]. Sigma phase was also reported in a more complex alloy system, $Al_{0.5}CoCrCuFeNiV_x$ [12]. On the other hand, no sigma phase was observed in $AlCoCrFeNiV_x$ or AlCrFeCoNiCuV alloys [23,24], and refractory HEAs tend to form solid solution phases when V is added [25–29]. In this work, we examine microstructure and mechanical properties of the $CoCrFeMnNiV_x$ alloys. Two main objectives are pursued: (i) to gain new understanding on the effect of alloying with V on phase content in the $CoCrFeMnNiV_x$ alloys; and (ii) to establish relationships between the microstructure and mechanical properties of the $CoCrFeMnNiV_x$ alloys.

2. Experimental procedures

The alloys with nominal compositions of CoCrFeMnNi, $CoCrFeMnNiV_{0.25}$, $CoCrFeMnNiV_{0.5}$, $CoCrFeMnNiV_{0.75}$ and CoCrFeMnNiV were produced by arc melting of the elements in a low-pressure, high-purity argon atmosphere inside a water-cooled copper cavity. The purities of the alloying elements were above 99.9 (at.)%. To ensure chemical homogeneity, the ingots were flipped over and re-melted a least 5 times. The produced ingots of the CoCrFeMnNi and CoCrFeMnNiV alloys had dimensions of about $6 \times 15 \times 60$ mm^3. Three other ingots had dimensions of about $6 \times 8 \times 45$ mm^3. The actual chemical compositions of the produced alloys are given in Table 2. The alloys were studied both in as-solidified state and after homogenization annealing. Annealing was carried out at 1000 °C for 24 h, in accord with regime used previously for CoCrFeMnNi and CoCrFeMnNiV alloys [13]. Prior to homogenization, the samples were sealed in vacuumed (10^{-2} torr) quartz tubes filled with titanium chips to prevent oxidation. After annealing, the tubes with the samples were removed from the furnace and were cooled down to room temperature due to heat exchange with surrounding air.

Microstructure and phase composition of the alloys were studied using X-ray diffraction (XRD), scanning electron microscopy (SEM) and transmission electron microscopy (TEM) techniques. The XRD analysis was performed using RIGAKU diffractometer and Cu Kα radiation. Samples for SEM observations were prepared by careful mechanical polishing. SEM investigations were performed utilizing Quanta 200 3D microscope equipped with energy-dispersive (EDS) detector. EBSD phase mapping was conducted using Nova NanoSEM 450 microscope equipped with Hikari EBSD detector. Samples for TEM analysis were prepared by conventional twin-jet electro-polishing of mechanically pre-thinned to 100 μm foils, in a mixture of 95% C_2H_5OH and 5% $HClO_4$ at the 27 V potential. TEM investigations were performed using JEOL JEM-2100 apparatus, equipped with an EDS detector, at an accelerating voltage of 200 kV.

Vickers microhardness, HV, was measured on polished cross-section surfaces using a 136-degree Vickers diamond pyramid under a 250 g load applied for 15 s. For microhardness measurements of individual phases, 100 g load was applied. Compressive tests were performed on rectangular specimens with dimensions of $7 \times 5 \times 5$ mm^3 using the Instron 5882 machine. The initial strain rate was 10^{-3} s^{-1}.

Table 1
Phases and their lattice parameters of the as-solidified $CoCrFeMnNiV_x$ alloys (x = 0; 0.25; 0.5; 0.75; 1), in accord to the XRD analysis.

Alloy	Lattice	Lattice parameters (Å)
CoCrFeMnNi	FCC	3.592 ± 0.002
$CoCrFeMnNiV_{0.25}$	FCC	3.597 ± 0.001
$CoCrFeMnNiV_{0.5}$	FCC	3.606 ± 0.001
	Tetragonal	a = 8.820 ± 0.005, c = 4.569 ± 0.003, c/a = 0.518
$CoCrFeMnNiV_{0.75}$	FCC	3.607 ± 0.002
	Tetragonal	a = 8.822 ± 0.004, c = 4.573 ± 0.002, c/a = 0.518
CoCrFeMnNiV	FCC	3.603 ± 0.002
	Tetragonal	a = 8.827 ± 0.004, c = 4.579 ± 0.002, c/a = 0.519

Table 2
Chemical composition of different structural constituents in as-solidified $CoCrFeMnNiV_x$ (x = 0; 0.25; 0.5; 0.75; 1) alloys, as determined by SEM-based EDS-analysis. Typical regions used for the analysis are shown in Fig. 2. Actual average chemical compositions of the alloys are also given.

Element (at.%) Constituent No	Designation	Co	Cr	Fe	Ni	Mn	V
		CoCrFeMnNi					
1	Dendrite	21.0	21.8	22.1	16.7	16.4	–
2	Interdendrite	16.8	17.0	16.7	22.9	26.6	–
	Alloy composition	19.9	20.6	20.1	20.0	19.5	–
		$CoCrFeMnNiV_{0.25}$					
1	Grains	19.3	20.0	19.6	19.5	17.0	4.6
	Alloy composition	19.3	20.0	19.6	19.5	17.0	4.6
		$CoCrFeMnNiV_{0.5}$					
1	Grains	18.5	18.3	18.8	18.5	17.2	8.8
2	Light particles	16.0	24.7	17.5	12.1	17.3	12.4
3	Dark particles	18.6	18.7	19.3	17.5	16.3	9.5
	Alloy composition	18.9	18.8	18.4	18.0	17.8	9.1
		$CoCrFeMnNiV_{0.75}$					
1	Grains	17.5	15.0	17.01	20.8	18.9	10.8
2	Light particles	16.6	21.4	19.1	13.3	15.3	14.1
3	Dark particles	18.0	15.1	17.7	20.6	17.8	10.8
	Alloy composition	17.2	17.5	18.2	17.2	16.5	13.3
		CoCrFeMnNiV					
1	Grain interiors	16.3	18.0	17.4	15.6	15.5	17.2
2	Grain boundary phase	16.1	12.6	15.5	22.7	19.1	14.0
	Alloy composition	16.2	17.0	17.2	16.5	16.1	17.0

3. Results

3.1. Microstructure of the as-solidified $CoCrFeMnNiV_x$ alloys

XRD patterns of the as-solidified alloys are shown in Fig. 1, and the identified phases with their lattice parameters are given in Table 1. Two large-intensity diffraction peaks, (200) and (311), and a very tiny (111) peak – all from an fcc crystal lattice, are found in the XRD pattern of CoCrFeMnNi, indicating presence of a strong crystallographic texture or large grains. A single fcc phase structure is also detected in the $CoCrFeMnNiV_{0.25}$ alloy. For $CoCrFeMnNiV_{0.5}$, five strong diffraction peaks from the fcc crystal lattice are identified within the studied 2-Theta range. In addition, weak peaks from a tetragonal crystal lattice are also recognized. Intensities of diffraction peaks from the tetragonal crystal lattice increase and intensities of the peaks from the fcc lattice decrease with an increase in the concentration of V in $CoCrFeMnNiV_{0.75}$ and CoCrFeMnNiV. The lattice parameter of the fcc phase continuously

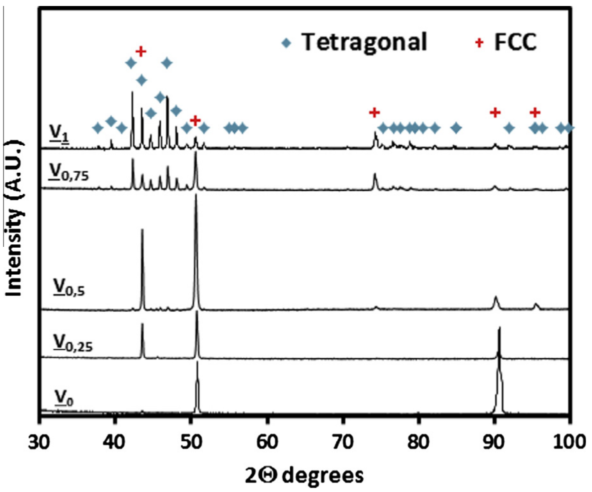

Fig. 1. XRD patterns of the CoCrFeMnNiV$_x$ (x = 0; 0.25; 0.5; 0.75; 1) alloys in the as-solidified state.

increases, from a = 3.592 Å to a = 3.606 Å, with an increase in the V content from x = 0 to x = 0.5 (0–9.1 at.%) and then retains almost the same for alloys with x = 0.5, 0.75 and 1.0 (i.e. V concentration ranging from 9.1 to 17.0 at.%) (Table 1). The lattice parameters of the tetragonal phase continuously increase from a = 8.820 Å and c = 4.569 Å to a = 8.827 Å and c = 4.579 Å when the V content increases from x = 0.5 to x = 1 (9.1–17.0 at.%), while the c/a ratio remains almost constant and equal to 0.518–0.519 in all three alloys (Table 1).

Typical microstructures of the CoCrFeMnNiV$_x$ alloys in as-solidified condition are shown in Fig. 2. The base CoCrFeMnNi alloy demonstrates a dendritic structure (Fig. 2a). The dendrite areas (light-gray ones, point 1) are slightly enriched (to 21–22%) with Co, Cr and Fe (Table 2) and contain about 16.4–16.7% of both Ni and Mn. Contrary, inter-dendritic areas (dark-gray ones, point 2) are enriched with Ni (22.9%) and Mn (26.6%) and contain about 16.7–17.0% of each of the remaining elements. The average grain size in this alloy is roughly estimated to be ~300 μm (Table 3), and the inter-dendrite spacing is about several tens of μm. The CoCrFeMnNiV$_{0.25}$ alloy is a single phase fcc structure consisting of elongated grains with an average length of 120 μm and an average width of 50 μm (Fig. 2b, Table 3). The chemical compositions of different grains are almost the same and correspond to the alloy composition.

The microstructure of the CoCrFeMnNiV$_{0.5}$ alloy consists of elongated grains of the fcc matrix (point 1, Fig. 2c), with an average grain length of 220 μm and an average width of 60 μm (Table 3). Irregularly-shaped bright-color particles (point 2,) with an average size of ~12 μm and volume fraction of 20% are observed both at grain boundaries and inside the grains. Inside some of the bright-color particles, smaller dark particles are found (point 3). The bright particles are enriched with Cr (about 24.7%) and V (about 12.4%) and depleted of Ni (about 12.1%) (Table 2). The matrix grains, on the other hand, are depleted of V (about 8.8%) and Cr (about 18.3%) and enriched with Ni (about 18.5%). The chemical composition of dark particles is very close to the composition of the matrix.

The microstructure of the CoCrFeMnNiV$_{0.75}$ alloy is similar to that of CoCrFeMnNiV$_{0.5}$, however the volume fraction of the bright particles (point 2, Fig. 2d, Table 3) increases to 37% while their average size is 16 μm. It should be noted that the particles have irregular shape. They are enriched with Cr (about 21.5%) and V (about 14.1%) and depleted of Ni (about 13.3%). Additionally, chains of smaller, round-shaped dark particles (point 3) with an average size of about 6 μm are present inside these bright particles. Chemical composition of the dark particles is very close to the chemical composition of matrix grains (point 1): both are enriched with Ni (about 20.8%) and depleted of Cr (about 15.0%) and V (about 10.8%). The average length and width of elongated grains of the matrix are equal to 500 μm and 100 μm, respectively (Table 3). A complex structure is observed in the CoCrFeMnNiV equiatomic alloy (Fig. 2e). It is composed of the matrix grains (point 1) and almost continuous network of a second phase at grain boundaries (point 2). The matrix grains are slightly enriched with Cr and V (18.0% and 17.2% respectively) and depleted of Ni (about 15.6%). The grain-boundary phase is enriched with Ni and Mn (about 22.7% and 19.1% respectively) (Table 2). The average matrix grain size (measured as the average distance between grain boundary phase particles) is about 20 μm (Table 3). Higher magnification images reveal that the matrix grains consist of two nano-lamellar phases (shown in a higher magnification insert): light matrix and dark elongated particles. EBSD phase mapping (Fig. 2f) reveals that the matrix phase has a tetragonal lattice (yellow color) and both the grain boundary phase and the particles inside the matrix have an fcc crystal structure (red color). The volume fraction of the matrix phase is estimated to be ~67% (Table 3).

Fine details of the microstructure of the as-solidified CoCrFeMnNiV$_x$ alloys were studied using TEM, and the results are given in Fig. 3 and Table 4. The crystal structure of phases was determined from the analysis of selected area electron diffraction patterns (SAEDPs), which are given together with respective bright-field TEM images. Slight variations of the chemical composition of the fcc phase are found in the V-free CoCrFeMnNi alloy (Fig. 3a, Table 4). From comparison of the chemical composition determined by SEM–EDS (Table 4) and TEM–EDS (Table 3) it can be deduced that point 1 in Fig. 3a belongs to a dendrite area, and point 2 – to an interdendrite area.

A single phase fcc structure with the composition close to the average alloy composition is found in the CoCrFeMnNiV$_{0.25}$ alloy (Fig. 3b, Table 4). Particles with a tetragonal crystal lattice enriched with Cr and V and depleted of Mn (Table 4) are identified as a sigma phase (point 2 in Fig. 3c) in the CoCeFeNiMnV$_{0.5}$ alloy. In the CoCrFeMnNiV$_{0.75}$ alloy (Fig. 3d), small particles with an fcc crystal lattice (point 3) are found inside larger particles with a tetragonal crystal lattice (point 2), which, in turn, are located inside the fcc matrix (point 1). The chemical composition of the fcc particles are almost identical to the composition of the fcc matrix (Table 3), but different from the composition of the particles with a tetragonal crystal structure, which are enriched with Cr and V and depleted of Ni. Finally, the microstructure of the grain interior region of the equimolar CoCrFeMnNiV alloy is shown in Fig. 3e. It is composed of the matrix phase with a tetragonal crystal structure and thus identified as a sigma phase, enriched with Cr and V, and depleted of Ni and Mn (Table 4). Inside the matrix, elongated particles with an fcc crystal structure are found. These particles are depleted of Cr and V and enriched with Ni and Mn. One can also note that the TEM-based EDS analysis (Table 4) of the sigma phase returns higher concentrations of Cr and V in comparison with the SEM-based analysis (Table 2) of the same condition. The difference could be roughly estimated as 2–3%. Probably a higher probe size used during SEM/EDS exited not only sigma phase regions, but also the nearest regions of the fcc phase depleted of Cr and V.

3.2. Microstructure of the CoCrFeMnNiV$_x$ alloys after annealing at 1000 °C

XRD patterns of the CoCrFeMnNiV$_x$ alloys after homogenization treatment are given in Fig. 4. Multiple diffraction peaks from fcc lattice are observed in the five-component CoCrFeMnNi alloy together with several weak peaks from Mn_2O_3 oxide. In the

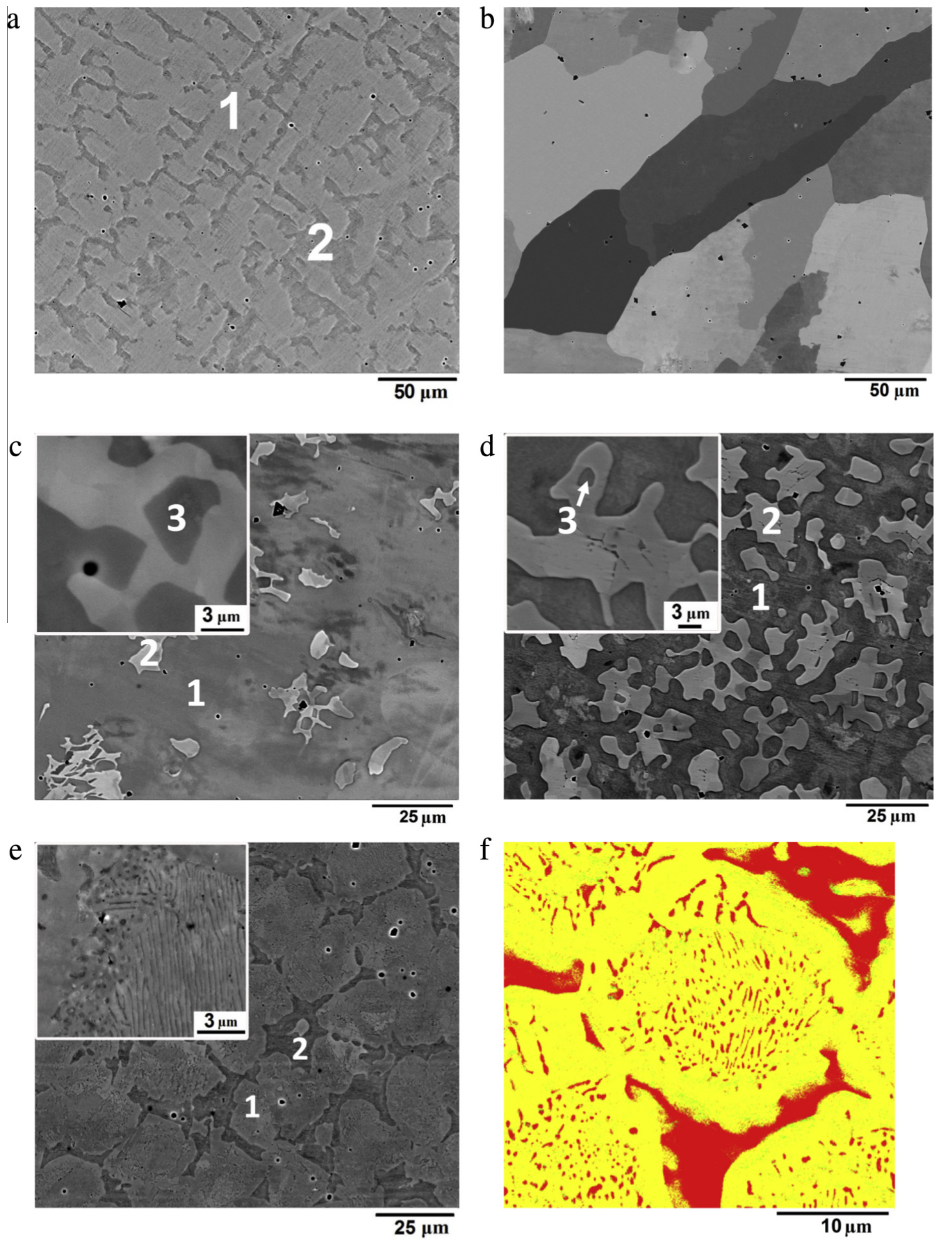

Fig. 2. Microstructure of CoCrFeMnNiV$_x$ (x = 0; 0.25; 0.5; 0.75; 1) alloys in as-solidified state, (a–e) SEM–BSE images and (f) EBSD phase map; (a) CoCrFeMnNi; (b) CoCrFeMnNiV$_{0.25}$; (c) CoCrFeMnNiV$_{0.5}$; (d) CoCrFeMnNiV$_{0.75}$; (e, f) CoCrFeMnNiV. The EBSD phase map shows the presence of a tetragonal sigma phase (yellow color) and an fcc phase (red color). The chemical composition of the numbered structural constituents is given in Table 2. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 3
Numerical information on the sizes of structural constituents of CoCrFeMnNiV$_x$ (x = 0; 0.25; 0.5; 0.75; 1) and their volume fraction in as-solidified condition and after annealing at 1000 °C as determined from SEM images.

	As-solidified			Annealed		
	FCC grain/particle size (μm)	Sigma particle/grain size (μm)	Sigma volume fraction (%)	FCC grain/particle size (μm)	Sigma particle/grain size (μm)	Sigma volume fraction (%)
CoCrFeMnNi	≈300	–	0	≈350	–	0
CoCrFeMnNiV$_{0.25}$	120	–	0	130	3	2.2
CoCrFeMnNiV$_{0.5}$	220	12	20	250	15/1.4[a]	28
CoCrFeMnNiV$_{0.75}$	≈500	16	37	≈500	22	48
CoCrFeMnNiV	–	20	67	12	≈35	72

[a] The average size of the coarse sigma phase particles is 15 μm, and the average thickness of lens-shaped particles is 1.4 μm.

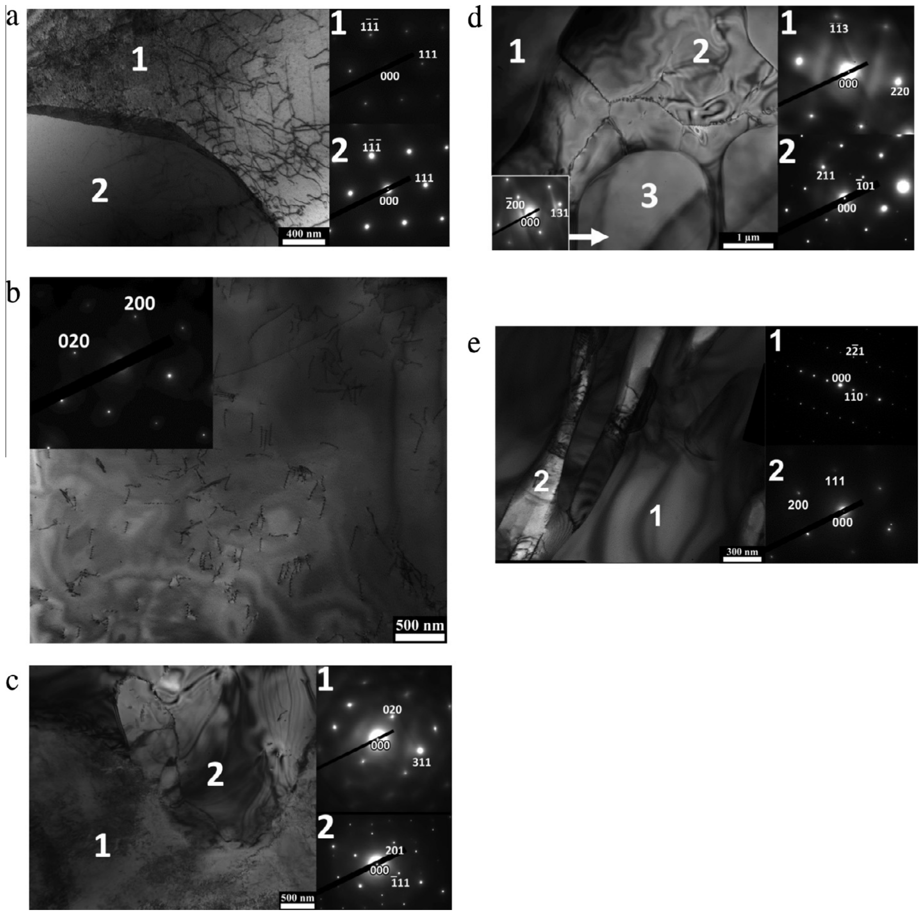

Fig. 3. Bright field TEM images of the microstructure of $CoCrFeMnNiV_x$ ($x = 0$; 0.25; 0.5; 0.75; 1) alloys and selected area electron diffraction patterns (SAEDs) from areas indicated by the numbers: (a) CoCrFeMnNi; (b) $CoCrFeMnNiV_{0.25}$; (c) $CoCrFeMnNiV_{0.5}$; (d) $CoCrFeMnNiV_{0.75}$; (e) CoCrFeMnNiV. The chemical compositions of the numbered structural constituents are given in Table 3.

Table 4
Crystal structure and chemical composition of different structural constituents of $CoCrFeMnNiV_x$ ($x = 0$; 0.25; 0.5; 0.75; 1) alloys in as-solidified state, as determined by TEM-based EDs-analysis. Typical regions of the analysis are shown in Fig. 3. For comparison, actual average chemical compositions of the alloys are also given.

Element (at.%) Constituent			Co	Cr	Fe	Ni	Mn	V
No	Designation	Lattice type						
			CoCrFeMnNi					
1	Dendrite	FCC	19.3	22.7	19.9	17.9	20.2	–
2	Interdendrite	FCC	17.3	20.2	16.9	20.0	25.7	–
	Alloy composition		19.9	20.6	20.1	20.0	19.5	–
			$CoCrFeMnNiV_{0.25}$					
1	Grains	FCC	19.7	21.6	21.2	17.3	15.7	4.6
	Alloy composition		19.3	20.0	19.6	19.5	17.0	4.6
			$CoCrFeMnNiV_{0.5}$					
1	Matrix	FCC	17.1	18.9	15.5	18.0	21.4	9.2
2	Particle	Tetragonal	15.6	26.6	17.2	10.5	17.3	12.7
	Alloy composition		18.9	18.8	18.4	18.0	17.8	9.1
			$CoCrFeMnNiV_{0.75}$					
1	Matrix	FCC	17.5	16.1	18.2	18.6	17.8	11.9
2	Particle	Tetragonal	16.9	21.5	17.3	12.2	15.2	16.8
3	Particle	FCC	17.5	15.7	17.4	18.8	18.4	12.3
	Alloy composition		17.2	17.5	18.2	17.2	16.5	13.3
			CoCrFeMnNiV					
1	Matrix	Tetragonal	16.3	19.1	15.4	16.2	14.0	18.9
2	Particle	FCC	18.0	14.1	17.3	17.6	19.6	13.4
	Alloy composition		16.2	17.0	17.2	16.5	16.1	17.0

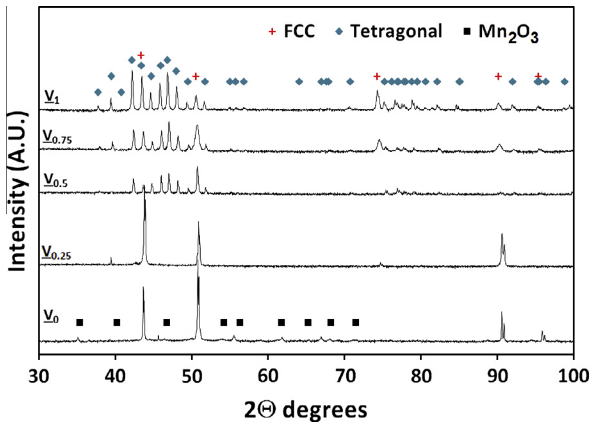

Fig. 4. XRD patterns of the $CoCrFeMnNiV_x$ ($x = 0$; 0.25; 0.5; 0.75; 1) alloys after homogenization treatment.

Table 5
Phases and their lattice parameters of the homogenized $CoCrFeMnNiV_x$ ($x = 0$; 0.25; 0.5; 0.75; 1) alloys calculated using of the respective XRD patterns.

Alloy	Lattice	Lattice parameter (Å)
CoCrFeMnNi	FCC	3.595 ± 0.001
$CoCrFeMnNiV_{0.25}$	FCC	3.594 ± 0.001
$CoCrFeMnNiV_{0.5}$	FCC	3.601 ± 0.001
	Tetragonal	a = 8.815 ± 0.004, c = 4.550 ± 0.002, c/a = 0.516
$CoCrFeMnNiV_{0.75}$	FCC	3.604 ± 0.002
	Tetragonal	a = 8.819 ± 0.004, c = 4.551 ± 0.002, c/a = 0.516
CoCrFeMnNiV	FCC	3.608 ± 0.002
	Tetragonal	a = 8.829 ± 0.005, c = 4.576 ± 0.003, c/a = 0.518

$CoCrFeMnNiV_{0.25}$ alloy, diffraction peaks from the fcc lattice are observed alongside with very weak peaks from a tetragonal lattice. In alloys with higher V contents, intensities of the diffraction peaks from the tetragonal phase increase and intensities of the peaks from the fcc lattice decrease relative to the respective as-solidified conditions (Fig. 1). The analysis of the XRD patterns, given in Table 5, shows a gradual increase in the lattice parameter of the fcc phase with an increase in the V content in all V-containing alloys. For example, the fcc lattice parameter increases from 3.594 Å in the $CoCrFeMnNiV_{0.25}$ alloy to 3.608 Å in the CoCrFeMnNiV alloy. The fcc lattice parameter of the CoCrFeMnNi alloy is 3.595 Å, i.e. almost equal (taking in account the error margin) to than that of the $CoCrFeMnNiV_{0.25}$ alloy. The lattice parameters of the tetragonal phase also increase with an increase in the V content, from a = 8.815 Å and c = 4.550 Å in the $CoCrFeMnNiV_{0.5}$ alloy to a = 8.829 Å and c = 4.576 Å in the CoCrFeMnNiV alloy (reflections from the tetragonal phase in the $CoCrFeMnNiV_{0.25}$ alloy are too weak to adequately calculate the lattice parameters). However, c/a ratio of the tetragonal phase remains almost unaffected with the change in the chemical composition and retains values of 0.516–0.518.

Microstructures of the homogenized alloys are shown in Fig. 5. It is clearly seen that annealing results in elimination of the dendritic structure in the CoCrFeMnNi alloy (Fig. 5a). After annealing, the alloy has a coarse-grained structure (average grain size of about 350 μm) (Table 3). The composition of grains is similar to the composition of the alloy (Table 6). Somewhat similar microstructure, but with finer average grain size of 130 μm, is observed in the $CoCrFeMnNiV_{0.25}$ alloy (Fig. 5b). However, small amount (about 2%) of bright particles (point 2, high magnification insert in Fig. 5b) enriched with Cr (32.4%) and V (6.2%) (Table 6) are found preferably at the grain boundaries. A complex structure is observed in the homogenized $CoCrFeMnNiV_{0.5}$ alloy (Fig. 5c). Two types of bright particles are found inside the darker matrix (point 1) with average grain size of 250 μm (Table 3): reasonably equiaxed ones, with the average size of 15 μm (point 2), and thin lens-shaped particles, with average length of 29 μm and thickness of 1.4 μm (point 3) (Table 3). The fact that lens-shaped particles have well-defined preferred orientations (they can be oriented in three different directions which are inclined by 60° to each other) implies the presence of crystallographic orientation relationships between the particles and the matrix. The chemical composition of two types of the particles is similar: they are enriched with Cr (29.7–29.9%) and V (12.0–12.4%) and depleted of Ni (10.7–10.8%) (Table 6). The matrix is in turn depleted in Cr (16.1%) and V (7.7%) and contains at least 19% of other elements. The volume fraction of the second-phase particles is estimated as 28%.

The microstructure of the annealed $CoCrFeMnNiV_{0.75}$ (Fig. 5d) is reasonably similar to that of the as-solidified alloy. It consists of bright-color matrix (point 1) with very large grains enriched with Ni (22.6%) (Table 6) and depleted of Cr (12.6%) and V (10.9%) and dark, irregularly shaped particles (point 2) enriched with Cr (22.6%) and V (16.0%). The average size of the dark particles is 22 μm, and their volume fraction is 48% (Table 3). Inside some dark particles, lighter ones (point 3, higher magnification insert in Fig. 5d) with the average size of 6 μm are observed. The chemical composition of the lighter particles is almost identical to the chemical composition of the matrix (Table 6). The microstructure of the homogenized condition of the equiatomic CoCrFeMnNiV alloy (Fig. 5c) appears similar to the microstructure of $CoCrFeMnNiV_{0.75}$. It consists of the matrix phase (point 1) with two different types of particles distributed inside. The first type is coarse, irregularly shaped particles (point 2) and the second type is fine equiaxed particles (point 3). The average grain size of matrix phase is ≈35 μm, and average size of particles is 12 μm (Table 3). However, unlike all the previous alloys, the matrix phase of CoCrFeMnNiV is sigma enriched with Cr (20.0%) and V (18.6%) and depleted with Ni (13.5%), and particles are fcc structures enriched with Ni (24.5–24.6%) (Table 6). The volume fraction of the particles is about 28% (Table 3).

TEM images of the annealed $CoCrFeMnNiV_x$ alloys are given in Fig. 6. TEM micrograph and corresponding SAEDP from the CoCrFeMnNi alloy demonstrate a FCC solid solution structure with composition reasonably close to the nominal (Table 7). In the alloy with small V addition, $CoCrFeMnNiV_{0.25}$, fine sigma-phase particles with a tetragonal crystal lattice enriched with Cr (37.1%) and V (7.7%) (point 2, Fig. 6b and Table 7) are observed at grain boundaries of the fcc matrix phase (point 2). In $CoCrFeMnNiV_{0.5}$, elongated sigma-phase particles containing large amounts of Cr (29.2%) and V (12.4%) (point 2 Fig. 6c, Table 7) are located inside the fcc matrix, which is enriched with Ni (20.4%) (point 1). The morphology of the sigma particles implies that there is specific orientation relationship between the sigma and fcc phases. Following orientation relationship have been obtained $(220)fcc \parallel (-110)\sigma$, $[-2-20]fcc \parallel [-1-13]\sigma$ and $(00-2)fcc \parallel (332)\sigma$, $[-2-20]fcc \parallel [-1-13]\sigma$ (Fig. 6d). A similar structure is observed in the $CoCrFeMnNiV_{0.75}$ alloy, but the sigma-phase particles are located inside the fcc matrix (point 1, Fig. 6e) and have complex irregular shape. They contain 24.5% of Cr and 16.5% of V (point 2, Table 7). In the CoCrFeMnNiV alloy, a mixture of fine particles of two phases is observed (Fig. 6f). The matrix phase (point 1) is a sigma phase with a tetragonal crystal lattice. It is enriched with Cr (20.7%) and V (19.1%) (Table 7). Particles of the fcc phase (point 2) often contain twins and are enriched with Ni (22.7%). In comparison with the as-solidified condition, the sigma phase after annealing contains more Cr and V. The difference is on average about 2–3% of both elements.

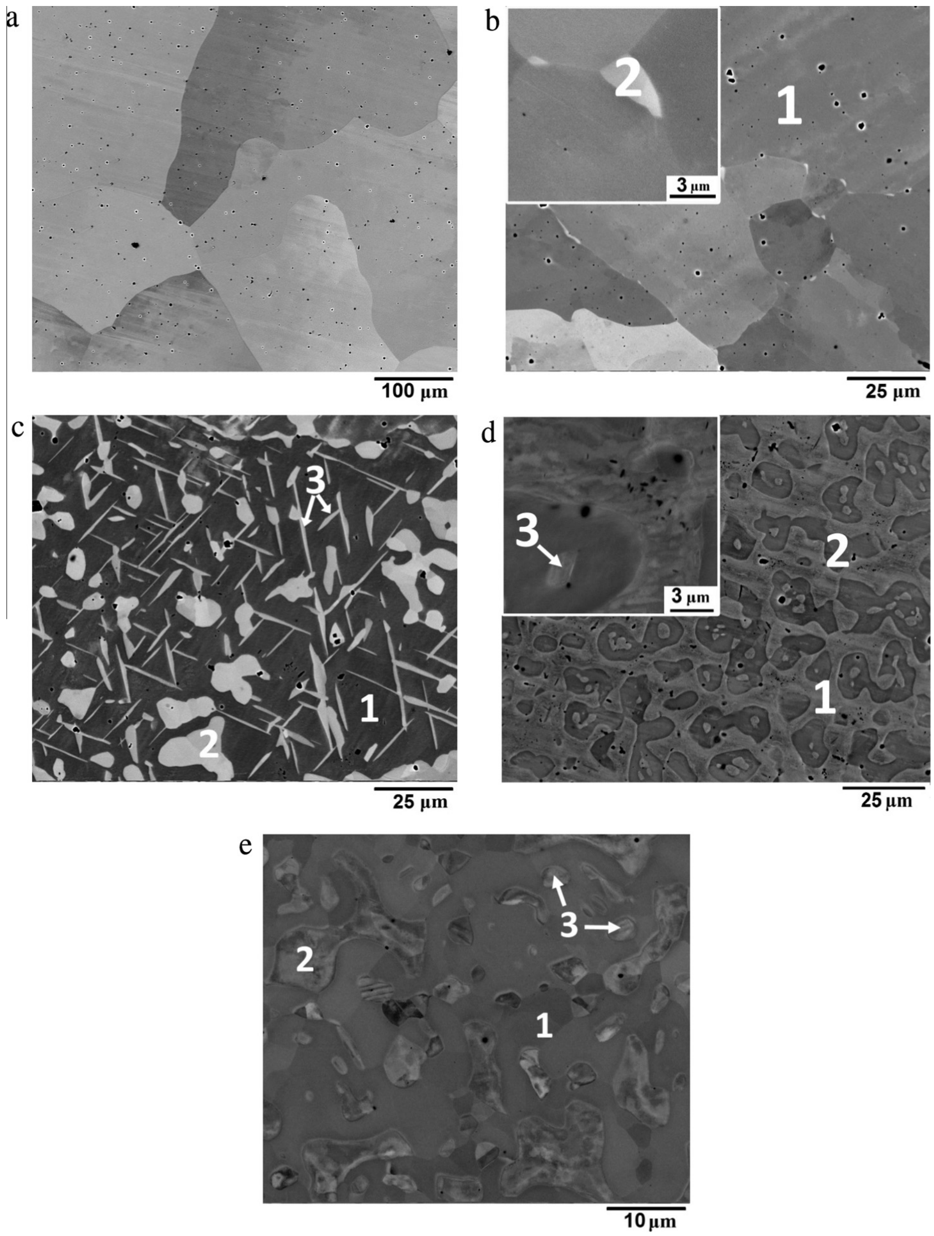

Fig. 5. SEM/BSE images of the microstructure of CoCrFeMnNiV$_x$ (x = 0; 0.25; 0.5; 0.75; 1) alloys after annealing at 1000 °C: (a) CoCrFeMnNi; (b) CoCrFeMnNiV$_{0.25}$; (c) CoCrFeMnNiV$_{0.5}$; (d) CoCrFeMnNiV$_{0.75}$; (e) CoCrFeMnNiV. The chemical compositions of the structural constituents identified in the figures are given in Table 6.

3.3. Mechanical properties of the CoCrFeMnNiV$_x$ alloys

Microhardness values of the CoCrFeMnNiV$_x$ alloys in as-solidified and annealed conditions are given in Table 8. The five-component equiatomic CoCrFeMnNi alloy in as-solidified state has microhardness of 144 HV. The CoCrFeMnNiV$_{0.25}$ alloy has slightly higher microhardness of 151 HV. A further increase in the V content results in a significant increase in microhardness. For example, CoCrFeMnNiV$_{0.5}$ has microhardness 186 HV and CoCrFeMnNiV has microhardness of 650 HV.

Annealing has pronounced effect on microhardness of the alloys. In the CoCrFeMnNi alloy microhardness decreases to 135 HV. A slight decrease in the microhardness value, down to 144 HV, is also found in CoCrFeMnNiV$_{0.25}$. However, a very significant increase in microhardness, to 275 HV, is observed in CoCrFeMnNiV$_{0.5}$. An increase in microhardness, from 342 HV in as-solidified state to 380 HV in the annealed state, is also found in the CoCrFeMnNiV$_{0.75}$ alloy. However, in the equiatomic CoCrFeMnNiV microhardness remains almost unaffected by annealing and is equal to 636 HV. In addition, microhardness of the matrix phase was measured using a smaller load of 100 g (Table 8). It shows similar tendencies with the overall hardness. It increases with an increase in the V concentration and decreases after annealing. It is slightly higher than the overall hardness in majority of the alloys, but in the CoCrFeMnNiV alloy hardness of the matrix phase is almost two times higher than the average hardness (1025–1125 HV and 636–650 HV respectively).

Mechanical properties of the alloys in the annealed condition were additionally studied using compressive testing. The stress–strain curves are shown in Fig. 7 and the mechanical properties such as yield strength, $\sigma_{0.2}$, peak strength, σ_u, and fracture strain, ε, are summarized in Table 9. Evidentially, V content has a very pronounced effect on the compressive behavior of the alloys. The V-free alloy has low yield strength of 230 MPa, and a very

Table 6
Chemical composition of different structural constituents of CoCrFeMnNiV$_x$ (x = 0; 0.25; 0.5; 0.75; 1) alloys after annealing at 1000 °C as determined by SEM-based EDS-analysis. Typical regions of analysis are shown in Fig. 5. For comparison actual chemical compositions of the alloys are also given.

Element (at.%) Constituent		Co	Cr	Fe	Ni	Mn	V
No	Designation						
		CoCrFeMnNi					
	Grains	19.6	20.5	19.9	20.9	19.1	–
	Alloy composition	19.9	20.6	20.1	20.0	19.5	–
		CoCrFeMnNiV$_{0.25}$					
1	Matrix	19.2	19.9	19.5	19.7	17.2	4.6
2	Particles	17.7	32.4	18.1	11.3	13.9	6.6
	Alloy composition	19.3	20.0	19.6	19.5	17.0	4.6
		CoCrFeMnNiV$_{0.5}$					
1	Matrix	19.2	16.2	19.2	19.6	18.3	7.7
2	Coarse light particles	16.8	27.9	17.3	10.8	14.8	12.4
3	Lens-shaped light particles	17.1	27.8	17.7	10.7	14.8	12.0
	Alloy composition	18.9	18.8	18.4	18.0	17.8	9.1
		CoCrFeMnNiV$_{0.75}$					
1	Matrix	17.4	12.6	18.2	22.6	18.3	10.9
2	Dark particles	16.8	22.6	17.9	11.9	14.8	16.0
3	Light particles	17.9	13.5	18.2	22.4	18.2	9.6
	Alloy composition	17.2	17.5	18.2	17.2	16.5	13.3
		CoCrFeMnNiV					
1	Grain interiors	16.7	20.0	16.6	13.5	14.6	18.6
2	Grain boundary phase	16.9	11.4	17.0	24.6	17.8	12.3
3	Particle	17.0	11.5	16.9	24.5	17.7	12.4
	Alloy composition	16.2	17.0	17.2	16.5	16.1	17.0

pronounced work hardening stage after yielding. It is very ductile and is compressed to 75% height reduction without fracture. The CoCrFeMnNiV$_{0.25}$ alloy exhibits very similar behavior, although its yield strength of 200 MPa is slightly lower than that of the V-free alloy. A further increase in the V content results in substantial strengthening: the yield strength of the CoCrFeMnNiV$_{0.5}$ alloy approaches 620 MPa, and the flow stress values at given strains are much higher than those of two previously discussed alloys. The alloy still has high compressive ductility of more than 75%, and a pronounced work hardening stage is observed on the early stages of deformation. However, at higher strains (30% and more) several drops of stress level are observed (Fig. 7). Even higher strengthening accompanied by deterioration of ductility is observed in alloys with higher V contents. For example, yield strength increases to 740 MPa in alloy with x = 0.75 and to 1660 MPa in alloy with x = 1. The CoCrFeMnNiV$_{0.75}$ alloy demonstrates some work hardening capacity and limited ductility of 7.8%. The peak strength value is 1325 MPa. The equiatomic CoCrFeMnNiV alloy has almost no plastic ductility (ε = 0.5) and it fractures at the maximum strength of 1845 MPa.

4. Discussion

4.1. Predicted phase composition of the CoCrFeMnNiV$_x$ alloys

Predicting phase equilibrium of compositionally complex (based on multiple principle components) HEAs is a complicated task [30]. The phase composition of the HEAs was initially suggested to be governed by the high entropy of mixing of the alloying elements [1]:

$$\Delta S_{\text{mix}} = -R\Sigma c_i \ln c_i \tag{1}$$

where R is the gas constant and c_i is the atomic fraction of element i. A high value of ΔS_{mix} was thought to be responsible for the predominant formation of solid solution phases, as the Gibbs free energy is reduced by the entropy term, $T\Delta S_{\text{mix}}$. Several additional criteria for predicting solid solution phase formation in HEAs were introduced recently [18–21]. In accord to Hume-Rothery rules for binary substitutional solid solutions [22], the solvent and solute should have: (i) small atomic size difference, (ii) the same valence, (iii) similar electronegativity and (iv) same crystal lattice. Also, thermodynamic parameters in addition to the entropy of mixing have to be considered. In current work we have calculated the following parameters for the studied CoCrFeMnNiV$_x$ alloys:

$$\delta r = 100\%\sqrt{\sum c_i(1 - r_i/\bar{r})^2} \tag{2}$$

$$\Delta H_{\text{mix}} = \sum 4\omega_{ij}c_i c_j \tag{3}$$

$$\Omega = T_m \Delta S_{\text{mix}}/|\Delta H_{\text{mix}}| \tag{4}$$

Here δr is atomic size difference [18], c_i and r_i are the atomic fraction and the atomic radius, respectively, of element i, $\bar{r} = \sum c_i r_i$ is the average atomic radius, ΔH_{mix} is the enthalpy of mixing [18], ω_{ij} is a concentration-dependent interaction parameter between elements i and j in a sub-regular solid solution model [31], Ω is a thermodynamic parameter specially introduced in [19], $T_m = \sum c_i T_{mi}$, and T_{mi} is the melting temperature of element i. It was found that solid solutions form in a number of HEAs for which δr < 6.2% and ΔH_{mix} is in the range from −20 kJ/mole to 5 kJ/mole, while intermetallic phases can be present in HEAs for which δr > 3% and ΔH_{mix} < 5 kJ/mole [13,18]. Only solid solution phases were found to form in many HEAs when the conditions $\Omega \geqslant 1.1$ and $\delta r \leqslant 6.6\%$ are met [19].

Some efforts were also made to develop specific criterions for formation of intermetallic phases in HEAs. Among them we have calculated the following ones:

$$\text{VEC} = \sum c_i \text{VEC}_i \tag{5}$$

$$\Delta\chi = \sqrt{\sum c_i(\chi_i - \bar{\chi})^2} \tag{6}$$

where VEC is the average valence electron concentration [21], VEC$_i$ is the valence electron concentration of element i, $\Delta\chi$ is Pauling electronegativity difference, χ_i is Pauling electronegativity of element i [32], and $\bar{\chi} = \sum c_i \chi_i$ is the average electronegativity. It was revealed that the HEAs containing elements of VB to VIIIB groups having VEC values between 6.88 and 7.84 are prone to σ phase formation either in the as-solidified state or during aging at suitable temperatures [33]. However, other authors found that at VEC in the range of 6.87–8.0 a mixture of fcc and bcc phases generally forms, at VEC ⩾ 8.0 fcc phases are stable and at VEC smaller than 6.87 bcc phases are formed [21]. Topologically close-packed phases were found to be stable in alloys with $\Delta\chi$ > 0.133, except for the alloys containing high concentrations of Al [27].

Formation of sigma phase in CoCrFeMnNi$_y$V$_x$ alloys (x, y = 0, 1) studied by Salishchev et al. [13] was correlated with large local distortions of the fcc lattice caused by V atoms, as well as VEC values in the range of 6.88–7.84. The following formula was used to estimate the average lattice distortion near element i caused by different atomic radii of the element i and its nearest neighbors:

$$\delta r_i = \frac{13}{12}\sum c_j \delta r_{ij} \tag{7}$$

where $\delta r_{ij} = 2(r_i - r_j)/(r_i + r_j)$ is atomic size difference, and c_j and r_j are respectively atomic fractions and atomic radii of element j ($i \neq j$). A critical relative expansion (or contraction) of the interatomic spacing that can change the coordination number near a solute atom has been calculated as $|\delta r_c|$ = 3.8% [34]. Therefore if δr_i value exceeds 3.8% one can anticipate instability of the solid

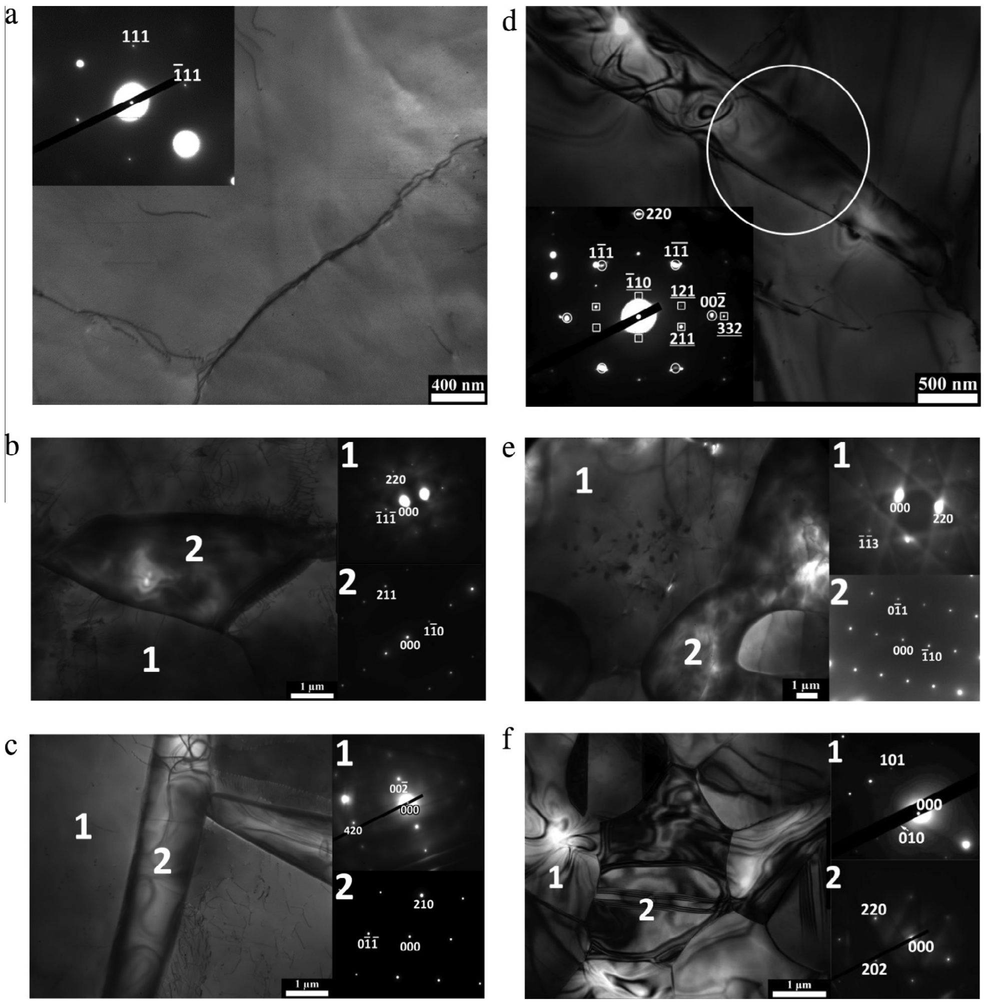

Fig. 6. Microstructure of CoCrFeMnNiV$_x$ (x = 0; 0.25; 0.5; 0.75; 1) alloys after annealing at 1000 °C, bright field TEM images with selected area electron diffraction patterns (SAEDs) from the areas indicated in figures: (a) CoCrFeMnNi; (b) CoCrFeMnNiV$_{0.25}$; (c, d) CoCrFeMnNiV$_{0.5}$; (e) CoCrFeMnNiV$_{0.75}$; (f) CoCrFeMnNiV. In fig d, indexed reflections from both the fcc matrix and sigma phase (underlined) are shown. Chemical compositions of the structural constituents identified in the figures are given in Table 7.

Table 7
Crystal structures and chemical compositions of different structural constituents of the CoCrFeMnNiV$_x$ (x = 0; 0.25; 0.5; 0.75; 1) alloys after annealing at 1000 °C as determined by TEM-based EDS-analysis. Typical regions of the analysis are shown in Fig. 6. For comparison actual chemical compositions of the alloys are also given.

Element (at.%) Constituent			Co	Cr	Fe	Ni	Mn	V
No	Designation	Lattice type						
			CoCrFeMnNi					
1	Grains	FCC	19.5	20.0	19.7	20.7	20.1	–
	Alloy composition		19.9	20.6	20.1	20.0	19.5	–
			CoCrFeMnNiV$_{0.25}$					
1	Matrix	FCC	18.7	21.3	19.0	18.0	18.5	4.6
2	Particle	Tetragonal	16.5	37.1	16.9	8.0	13.8	7.7
	Alloy composition		19.3	20.0	19.6	19.5	17.0	4.6
			CoCrFeMnNiV$_{0.5}$					
1	Matrix	FCC	18.6	17.1	19.1	17.9	19.1	8.2
2	Particle	Tetragonal	17.3	29.2	17.7	8.9	14.6	12.4
	Alloy composition		18.9	18.8	18.4	18.0	17.8	9.1
			CoCrFeMnNiV$_{0.75}$					
1	Matrix	FCC	17.3	13.5	18.7	20.4	19.5	10.6
2	Particle	Tetragonal	16.7	24.0	17.4	10.4	15.1	16.5
	Alloy composition		17.2	17.5	18.2	17.2	16.5	13.3
			CoCrFeMnNiV					
1	Matrix	Tetragonal	16.5	20.7	16.4	12.6	14.7	19.1
2	Particle	FCC	17.0	11.9	17.1	22.7	18.5	12.8
	Alloy composition		16.2	17.0	17.2	16.5	16.1	17.0

Table 8
Microhardness of the $CoCrFeMnNiV_x$ (x = 0; 0.25; 0.5; 0.75; 1) alloys in as-solidifies state and after annealing at 1000 °C.

	As-solidified		Annealed	
	Overall hardness (HV)[a]	Matrix phase hardness (HV)[b]	Overall hardness (HV)[a]	Matrix phase hardness (HV)[b]
CoCrFeMnNi	144 ± 3	161 ± 12	135 ± 2	159 ± 10
$CoCrFeMnNiV_{0.25}$	151 ± 5	165 ± 5	144 ± 5	153 ± 7
$CoCrFeMnNiV_{0.5}$	186 ± 12	199 ± 9	275 ± 7	315 ± 37
$CoCrFeMnNiV_{0.75}$	342 ± 13	378 ± 36	380 ± 14	474 ± 35
CoCrFeMnNiV	650 ± 27	1025 ± 91	636 ± 23	1125 ± 106

[a] Measured under 300 g load.
[b] Measured under 100 g load.

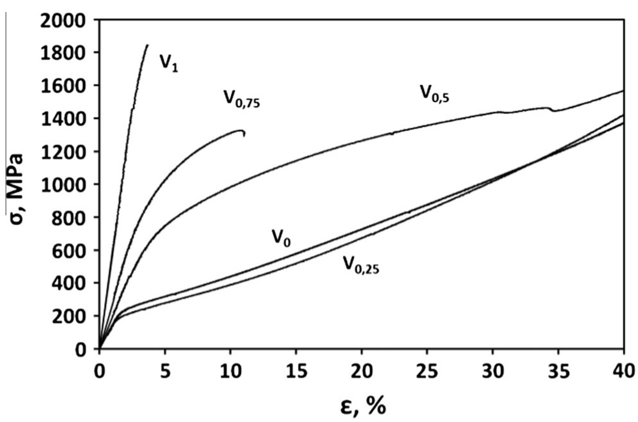

Fig. 7. Stress–strain curves obtained during compressive testing of the homogenized $CoCrFeMnNiV_x$ (x = 0; 0.25; 0.5; 0.75; 1) alloys.

Table 9
Compression yield strength, $\sigma_{0.2}$, fracture strength, σ_u, and fracture strain, ε, of the $CoCrFeMnNiV_x$ (x = 0; 0.25; 0.5; 0.75; 1) alloys after annealing at 1000 °C.

Alloy	$\dot{\varepsilon}$ (s^{-1})	$\sigma_{0.2}$ (MPa)	σ_u (MPa)	ε (%)
CoCrFeMnNi	10^{-3}	230	Not fractured	>75
$CoCrFeMnNiV_{0.25}$	10^{-3}	200	Not fractured	>75
$CoCrFeMnNiV_{0.5}$	10^{-3}	620	Not fractured	>75
$CoCrFeMnNiV_{0.75}$	10^{-3}	740	1325	7.8
CoCrFeMnNiV	10^{-3}	1660	1845	0.5

solution lattice near element i. In the $CoCrFeMnNi_yV_x$ (x, y = 0, 1) high distortions δr_V of 5.6–5.7% were thought to be the reason for sigma phase formation in V-containing alloys with VEC in the range of 6.88–7.84 [13].

All the mentioned parameters were calculated for the studied $CoCrFeMnNiV_x$ alloys using Eqs. (1)–(7) and the calculations are summarized in Table 10. It should be noted that only the lattice distortion values near V atoms, δr_V, are given in the table, because δr_i near other elements are considerably (more than an order of magnitude) smaller than the critical value of 3.8% and will not destabilize the solid solution crystal lattice.

Analysis of the parameters listed in Table 10 can provide some suggestions about phase stability of the studied alloys. A gradual increase in ΔS_{mix} with an increase in the V content in the $CoCrFeMnNiV_x$ alloys should benefit formation of solid solution phases. All five alloys have $\delta r \leqslant 3.0\%$, which also does not favor formation of intermetallic phases [18]. The ΔH_{mix} and Ω values of all alloys also fall in the range of solid solution phase formation [18,19]. On the other hand, the $\Delta\chi$ criterion [32] predicts formation of topologically close packed phases in all studied alloys, including CoCrFeMnNi. Thus none of the above-mentioned criteria, used individually, correctly predicts the vanadium-induced phase evolution in the studied alloy system. At the same time, the VEC criterion [21] correctly predicts a single fcc phase in CoCrFeMnNi, for which VEC = 8.0, and almost 100% of the fcc phase in $CoCrFeMnNiV_{0.25}$, for which VEC = 7.85. The VEC values for three other alloys, $CoCrFeMnNiV_{0.5}$, $CoCrFeMnNiV_{0.75}$, and CoCrFeMnNiV are 7.70, 7.61 and 7.50, respectively. These values fall in the VEC range from 7.84 to 6.88, where transformation is expected of the fcc phase to a bcc phase [21] or sigma phase [33]. The VEC criterion also correctly predicts that the volume fraction of the newly formed phase higher in the alloys with smaller values of VEC. The only drawback of the VEC criterion is that it does not recognize whether a bcc or sigma phase should form. This problem has already been pointed out in our previous paper [13] where an additional parameter, δr_i, was introduced. It was suggested that the sigma phase will form instead of the bcc phase if, in addition to the VEC criterion, δr_i exceeds the critical value of 3.8%. Otherwise, a bcc phase will form instead. This our suggestion is nicely supported by the current results for the $CoCrFeNiMnV_x$ alloys. Indeed the δr_V value is higher than the critical value of 3.8% in all the studied alloys containing V, which supports formation of the sigma phase. On the other hand, sigma phase was also reported in V-free alloys containing high amount of Cr instead, for example, in $CoCr_2FeNi$ [33]. These alloys have VEC in the range from 7.84 to 6.88, but the lattice distortions near any of the constitutive elements are below 3.8%, which may indicate that δr_i criterion does not work for the V-free alloys. At the same time, large shear modulus distortions have been reported near Cr atoms in these alloys [13], which could also result in instability of the solid solution phase. It can therefore be suggested that the combination of appropriate values of VEC, δr_i and δG_{Cr} should be considered for prediction of the formation of sigma phase in Cr and V containing HEAs. Additional work is required to validate this suggestion.

4.2. Effect of V content on the chemical composition of individual phases and their volume fraction in the $CoCrFeMnNiV_x$ alloys

The main effect of alloying a single-phase fcc CoCrFeMnNi alloy with V is formation of intermetallic sigma phase. According to the results of microstructural analysis, the volume fraction of sigma phase increases with an increase in V concentration. The chemical compositions of the fcc and sigma phases also change due to increasing the V content in the $CoCrFeMnNiV_x$ alloys (Tables 2, 4, 6 and 7). For example, when the V concentration is increased, the

Table 10
Calculated parameters ΔS_{mix}, δr, ΔH_{mix}, Ω, VEC, $\Delta\chi$ and δr_V for studied $CoCrFeMnNiV_x$ alloys.

Alloy	ΔS_{mix} (J/mole K)	δr (%)	ΔH_{mix} (kJ/mole)	Ω	VEC	$\Delta\chi$	δr_V
CoCrFeMnNi	13.38	1.1	−4.16	5.79	8.00	0.1384	–
$CoCrFeMnNiV_{0.25}$	14.36	1.8	−5.35	4.89	7.85	0.1381	0.063
$CoCrFeMnNiV_{0.5}$	14.69	2.1	−6.20	4.35	7.7	0.1376	0.061
$CoCrFeMnNiV_{0.75}$	14.85	2.4	−6.95	3.96	7.61	0.1369	0.058
CoCrFeMnNiV	14.9	2.6	−7.5	3.68	7.50	0.1361	0.056

concentrations of Co, Fe, Ni and Mn in the fcc phase decrease proportionally with a decrease in the average concentrations of these elements in the alloys. At the same time the concentrations of Cr and V in the fcc phase are reasonably close to the concentrations of these elements in the $CoCrFeMnNiV_{0.25}$ and $CoCrFeMnNiV_{0.5}$ alloys; while in the $CoCrFeMnNiV_{0.75}$ and CoCrFeMnNiV alloys the amounts of Cr and V in the fcc phase are considerably smaller than the respective alloy concentrations (Table 7). It is interesting to note that the total amount of Cr and V in the fcc phase in all alloys containing sigma phase tends to be nearly constant. For example, in the $CoCrFeMnNiV_{0.25}$ alloy in the annealed state the sum of atomic concentrations of Cr and V is 24.9%, and in the CoCrFeMnNiV alloy –24.7% (Table 7). Slightly higher amounts of Cr + V, about 26–28% (Table 5), are present in the fcc phase in as-solidified condition, which is probably due to non-equilibrium character of the microstructure. One can suggest that there is a maximum equilibrium solubility limit of Cr + V in the fcc phase of the $CoCrFeMnNiV_x$ alloys, apparently controlled by the sigma phase solvus curve. If the amount of Cr and V exceeds the solubility limit, formation of sigma phase occurs. To prove this suggestion, the volume fraction of sigma phase is plotted against the sum of the atomic concentrations of Cr and V in the studied $CoCrFeMnNiV_x$ alloys (Fig. 8a).

There is a threshold concentration of ~24% for Cr + V below which sigma phase does not form. Above this threshold, a linear dependence between the concentration of Cr + V and the volume fraction of the sigma phase is observed. It should be noted that the critical value of 24% of (Cr + V) is reasonably close to the maximum experimentally observed concentration of Cr and V in the fcc phase in the annealed condition. Fig. 8a also predicts about 38% of (Cr + V) to produce 100% of the sigma phase in the studied alloy system. It is interesting to note that the concentration of Cr + V in the sigma phase of the annealed specimens approximately corresponds to this upper limit (Tables 6 and 7), which may indicate that this concentration corresponds to the maximum solubility of other elements in the sigma phase. Therefore the dependences shown in Fig. 8a can provide insight into the phase content in the annealed $CoCrFeMnNiV_x$ alloys. The volume fraction of the sigma phase is clearly controlled by the concentration of Cr + V in the alloy and the constrained concentrations of these elements in the fcc and sigma phases, in accordance with the lever rule. Thereby, the phase composition of the $CoCrFeMnNiV_x$ alloys can be precisely controlled by altering their chemical compositions. Apparently, the maximum solubility of Cr + V in the fcc solid solution and, probably, the equilibrium concentrations of Cr and V in sigma phase are expected to be temperature dependent and thus some deviations from the dependences shown in Fig. 8a are expected due to sluggish diffusion of elements in the HEAs [35] if other annealing temperatures or non-equilibrium cooling conditions are used. For example, deviations of the experimentally observed concentrations of Cr + V in the fcc and sigma phases of the as-solidified alloys (Tables 2 and 4) from the critical (Cr + V) values of 24% and 38%, respectively, determined from the respective plot in Fig. 8a, are most probably due to non-equilibrium phase compositions after solidification.

Another factor, which can influence the phase composition of the $CoCrFeMnNiV_x$ alloys, is the ratio between Cr and V. The total concentration of Cr + V in sigma phase is clearly dependent on the V content in the alloys. For example, it decreases from 44.8% in the $CoFeFeNiMnV_{0.25}$ alloy to 39.8% in the CoCrFeMnNiV alloy (Table 7). An increase in the V concentration in the studied alloys results in partial replacement of Cr with V in sigma phase, which is expected as these element share the same sites in the tetragonal lattice of sigma phase [36]. When the volume fraction of sigma phase increases from ~2% annealed $CoFeFeNiMnV_{0.25}$ to ~67% in annealed CoFeFeNiMnV, the concentration of V in sigma phase increases from 7.7% to 19.1% and that of Cr decreases from 37.1% to 20.7%, respectively (Table 7). One can therefore suggest that V is more effective than Cr in stabilization of sigma phase. Analysis of binary phase diagrams supports our suggestion of higher efficiency of V (relative to Cr) as a sigma-forming element: (i) composition ranges for sigma phase in V-containing binary systems are wider or/and shifted toward smaller V concentrations in comparison with similar Cr-containing systems: 30–65% V in V–Fe and 45–50% in Cr–Fe system, 45–70% V in V–Co and 50–65% Cr in Cr–Co, 10–27% V in V–Mn and 20–25% Cr in Cr–Mn [30]; (ii) V forms a stable sigma phase with Ni whereas Cr does not [37]. The fact that V tends to form sigma phase with Ni is clearly reflected by an increase in Ni concentration in the sigma phase with an increase in the V content in the studied alloys (Tables 2, 4, 6 and 7). The concentrations of other elements, namely Co, Fe and Mn, in sigma phase are consistent with the change of the average composition of the alloys. Therefore, one can suggest that V is indeed more efficient for the formation of sigma phase and

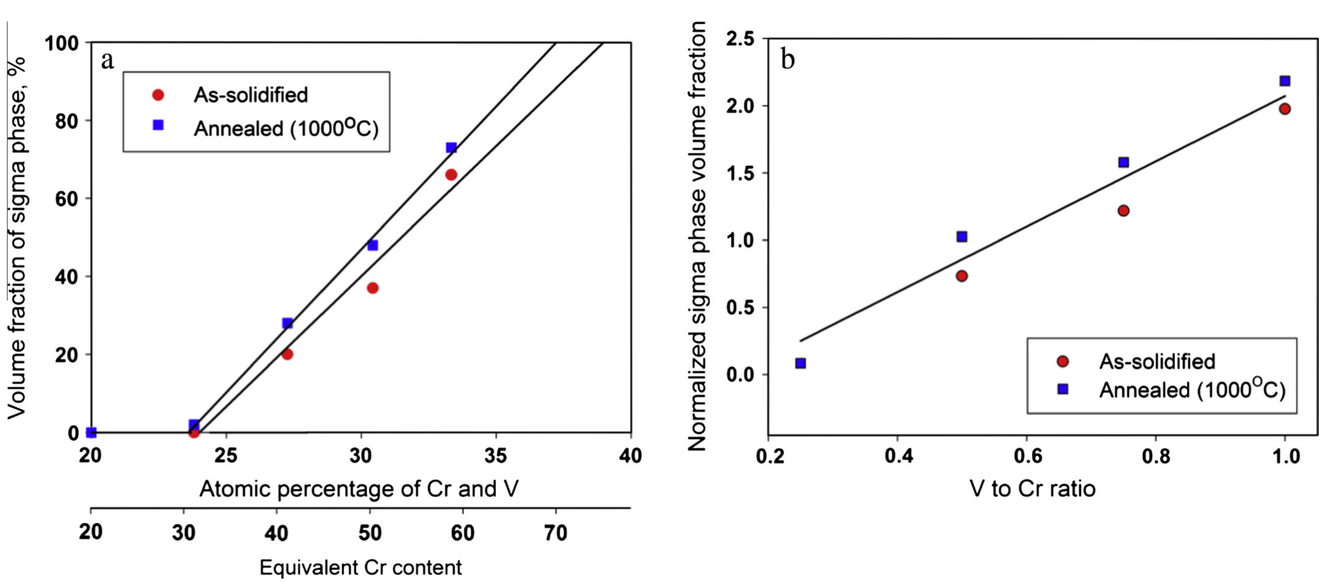

Fig. 8. The dependence of the volume fraction of sigma phase in the $CoCrFeMnNiV_x$ alloys on (a) total atomic concentration of Cr + V and equivalent Cr content and (b) normalized sigma phase volume fraction versus V to Cr ratio.

thus less amount of Cr and V together are observed in sigma phase in the $CoCrFeMnNiV_x$ alloys with increasing V content.

To check this assumption, we plotted a normalized sigma phase volume fraction against atomic ratio of V to Cr (Fig. 8b). For the purpose of normalization, a ratio between the volume fraction of sigma phase in a $CoCrFeMnNiV_x$ alloy and the atomic concentration of Cr + V in this alloy was used. The volume fraction of sigma phase was shown (Fig. 8a) to be linearly dependent on the Cr + V concentration in the case if sigma phase is present. Thus, the observed linear increase in the normalized volume fraction of the sigma phase is attributed to the increase in the V/Cr ratio (Fig. 8b). This proves that V has a positive effect on sigma phase formation. The slope of the dependence is 2.55, which means that the effect of V on the sigma phase formation in the studied alloys is 2.55 times higher than the effect of the same amount of Cr. In terms of "equivalent Cr concentration" [38,39], established for prediction of sigma phase in austenitic stainless steels, the following equation can be written:

$$Cr_{eq} = Cr + 2.55\,V \tag{8}$$

where Cr and V are atomic percentages of Cr and V respectively. Here the coefficient of 2.55 is noticeably higher than the value of 2.02 reported for stainless steels [38]. The difference in the coefficient can be attributed to the different concentrations of V and other constitutive elements in the studied $CoCrFeMnNiV_x$ alloys and steels (in [38], the concentration of V did not exceed 5%). The "effective Cr concentration" approach can be applied to other sigma phase forming elements, like Mo [9], after establishing the efficiency coefficients for the respective elements in Eq. (8). To summarize, using the dependence of Fig. 8a and "effective Cr concentration" concept, one can predict the volume fraction of the sigma phase in the $CoFeNiMnCr_xV_y$ system and design alloys with required phase composition. However, additions of other elements can also have significant effect on the phase composition of the alloys. For example, in a series of $AlCoCrFeNiV_x$ alloys (x = 0, 0.2, 0.5, 0.8, 1) no sigma phase was observed, probably due to the presence of Al, which restricts formation of sigma phase [23].

4.3. Morphology of the microstructure of the $CoCrFeMnNiV_x$ alloys

4.3.1. As-solidified condition

An increase in the V content in the $CoCrFeMnNiV_x$ alloys not only increases the volume fraction of sigma phase, as it was pointed out in previous section, but also results in noticeable changes in the microstructure, in both as-solidified and annealed conditions. In the as-solidified state, the CoCrFeMnNi and $CoCrFeMnNiV_{0.25}$ alloys are single-phase fcc solid solutions, although in the CoCrFeMnNi alloy segregation of high melting point elements such as Cr, Co and Fe and low melting point elements such as Mn and Ni results in the formation of a dendritic structure. Previous experimental studies of the microstructure of the CoCrFeMnNi alloy in as-cast condition [13,17] have also demonstrated dendritic segregations. Analysis of corresponding binary phase diagrams [37] demonstrates that (i) Co–Fe, Co–Ni, Fe–Ni, Fe–Mn, and Mn–Ni have unlimited mutual solubility in the fcc crystal lattice; and (ii) alloying of Fe with Co, Mn and Ni stabilizes fcc phase, while alloying with Cr and V stabilizes bcc phase. Thermodynamic modeling of the CoCrFeMnNi alloy by Zhang et al. [40] has shown that a single fcc phase is formed during solidification and is stable at temperatures at least down to 600 °C. Thus our results are in good agreement with both experimental data by other authors and thermodynamic predictions. In the $CoCrFeMnNiV_{0.25}$ alloy no segregation is found in the as-solidified state, probably due to either smaller size of the produced ingot or homogenizing effect of V addition, as it was described in [23]. It is also likely that an addition of V decreases the temperature range of solidification thus reducing microsegregation. Therefore, both CoCrFeMnNi and $CoCrFeMnNiV_{0.25}$ solidify similarly through the formation of a single fcc phase.

The as-solidified microstructure of the $CoCrFeMnNiV_{0.5}$, $CoCrFeMnNiV_{0.75}$ and CoCrFeMnNiV alloys consists of two phases, fcc and sigma. Formation of the sigma phase in the Cr and V-containing alloys is rather expected. Sigma phase is generally a product of solid state reactions and in steels it forms preferably from ferrite (a bcc phase) during aging at temperatures below 1000 °C, but it can also form directly from austenite (a fcc phase) [41]. It is also known that in V-containing systems sigma phase can form at much higher temperatures [37]. In a number of the alloying systems that do not contain Cr, namely in Co–V and Ni–V binary systems [37] and Co–Fe–V and Fe–Ni–V ternary systems [42,43], sigma phase forms from liquid. However, formation of the sigma phase directly from liquid phase is unlikely in the studied alloys that contain more Cr than V. Taking into account that Cr and V are bcc-stabilizing elements in austenitic steels, including Fe–Mn–Ni alloys, [37,41], one can suggest that at high temperatures, just below the melting point, the $CoCrFeMnNiV_{0.5}$, $CoCrFeMnNiV_{0.75}$ and CoCrFeMnNiV alloys have a duplex structure consisting of fcc and bcc phases, and the volume fraction of the bcc phase increases with an increase in the Cr + V content. Distinctive morphology of the sigma phase and the absence of the bcc phase in these alloys at room temperature suggest that the high-temperature bcc phase transforms to sigma and fcc phases by eutectoid-like reaction.

Indeed, irregular-shaped sigma-phase particles containing fcc particles are present inside the fcc matrix of the as-solidified $CoCrFeMnNiV_{0.5}$, $CoCrFeMnNiV_{0.75}$ alloys. These clusters of the sigma and fcc particles likely form from the high temperature bcc phase during cooling. At the same time, the as-solidified CoCrFeMnNiV alloy consists of grains with eutectoid-like structure, and an fcc grain boundary phase. The eutectoid-like structure is a mixture of sigma and fcc lamellar particles, with a higher volume fraction of sigma. Such a microstructure suggests that the bcc phase solidified first in this alloy and was enriched with Cr and V. The remaining liquid between the bcc grains became enriched with Mn and Ni and solidified in the fcc phase. During cooling in solid state, the bcc grains completely transformed to the mixture of the sigma and fcc phases. The sizes of the eutectoid-like clusters (in $CoCrFeMnNiV_{0.5}$, $CoCrFeMnNiV_{0.75}$) and grains (in CoCrFeMnNiV) containing the sigma and fcc particles indicate that the volume fraction of the high-temperature bcc phase increases with an increase in the V content from ~30% in $CoCrFeMnNiV_{0.5}$ to ~80% in CoCrFeMnNiV, which resulted in the formation of, respectively, ~20% and 67% of the sigma phase during the solid-state transformation. Similar bcc → fcc + sigma phase transformation is observed in duplex steels [44].

4.3.2. Annealed condition

The effect of annealing at 1000 °C on the microstructure of the studied $CoCrFeMnNiV_x$ alloys depends on the V content. In the CoCrFeMnNi alloy dendritic segregations observed in the as-solidified state are eliminated upon annealing and coarse granular structure with chemical composition of grains close to nominal composition of the alloy is formed. This result indicates that diffusivity of the alloying elements at 1000 °C is sufficient to homogenize the alloy within 24 h holding. The process-controlled diffusion coefficient, D, can be estimated using formula:

$$D = L^2/2t \tag{9}$$

where L is the average inter-dendrite spacing and t is the annealing time resulting in a homogeneous distribution of the alloying elements in the fcc phase. Using $L \approx 10$ μm and t = 24 h, D is estimated to be $D = 5.8 \times 10^{-16}$ m²/s (or higher, if the homogenization process is completed for less than 24 h). This estimated value can be compared with diffusion parameters for constitutive elements in the CoCrFeMnNi alloy given in [35]. At 1000 °C, the diffusion coefficients of the alloying elements are in the range from $\approx 1.8 \times 10^{-16}$ m^2/s for Ni (the slowest element) to $\approx 4.8 \times 10^{-16}$ m^2/s for Mn and $\approx 5 \times 10^{-16}$ m^2/s for Cr (the fastest element). Taking into account that the inter-dendritic areas of the CoCrFeNiMn alloy are enriched with Ni and Mn (Table 2), one can suggest that homogenization process of the CoCrFeNiMn alloy in controlled mainly by diffusion of these atoms. Using the diffusion coefficients reported in [35], the average diffusion distances for Ni and Mn during 24-h annealing at 1000 °C are estimated using Eq. (9) to be 5.6 μm and 9.1 μm, respectively.

About 2.2% of the sigma phase, in the form of fine particles located at grain boundaries, is observed in the CoCrFeMnNiV$_{0.25}$ alloy after annealing. This result may indicate that the alloy contains two equilibrium phases, fcc and bcc, at 1000 °C and during cooling from 1000 °C, the bcc phase transforms to sigma. The bcc phase in this alloy is likely the second phase to solidify in equilibrium condition. However, during rapid solidification after arc melting Cr and V are almost homogeneously distributed in the fcc phase and the remaining liquid (at the end of solidification) contains insufficient amount of V and Cr to nucleate the bcc phase. As Cr and V atoms cause significant distortions in the fcc solid solution (Table 10, [13]) they are probably forced out to the grain boundaries during long-term annealing to reduce stresses. Thus the concentrations of Cr and V at grain boundaries become sufficient to form the bcc phase, which transforms to the sigma phase during cooling. Another explanation of the sigma phase formation after annealing is heterogeneous precipitation of the sigma particles directly from the super-saturated fcc phase at temperatures below the sigma-phase solvus temperature, but high enough for diffusion. Unfortunately, the phase diagram for the CoCrFeMnNiV$_{0.25}$ alloy, as well as for the other V-containing alloys studied in this work, is currently unavailable, and an additional study is required to identify the mechanism of the sigma phase formation in this alloy.

An increase in the volume fraction of the sigma phase by ~5–11% (relative to the as-solidified state) is observed in all other V-containing alloys after annealing. This increase is most probably due to excessive contents of Cr and V in the fcc solid solution phase after non-equilibrium solidification, as it was discussed in previous section. Annealing at 1000 °C equilibrates the alloys, and an increase in the sigma volume fraction after annealing can be a result of (a) an increasing volume fraction of the bcc phase (if 1000 °C corresponds to the fcc + bcc phase filed) and/or (b) additional precipitation from a super-saturated fcc solid solution during slower cooling after annealing (if 1000 °C corresponds to fcc + sigma phase field and the solubility of Cr + V in the fcc phase rapidly decreases with a decrease in temperature). The experimental results show noticeable coarsening of the eutectoid-forming phases (sigma and fcc particles) in these three alloys, CoCrFeMnNiV$_{0.5}$, CoCrFeMnNiV$_{0.75}$, and CoCrFeMnNiV, after annealing, which suggests that the annealing temperature corresponds to the fcc + sigma phase field in these alloys. Therefore, an additional sigma phase likely precipitates from the super-saturated fcc phase during cooling after the annealing treatment. The different morphology of the sigma phase particles additionally formed in the annealed CoCrFeMnNiV$_{0.5}$ alloy from the particles observed in the as-solidified alloy supports this suggestion. Indeed, fine lens-shaped sigma particles precipitate inside the fcc grains, while the volume fraction of coarse sigma-phase particles present inside the eutectoid-like regions does not change after annealing of this alloy. In the CoCrFeMnNiV$_{0.75}$, and CoCrFeMnNiV alloys, which have higher volume fractions of the sigma-phase particles/grains after solidification, precipitation of an additional sigma phase from the fcc phase likely occurs at the fcc/sigma interfaces thus accelerating the sigma phase coarsening.

A lens-shaped morphology of sigma phase particles precipitated from the fcc phase in CoCrFeMnNiV$_{0.5}$ is an unusual feature, that requires more attention. A similar morphology of the sigma phase was also observed in the Fe–22Cr–21Ni–6Mo superaustenitic stainless steel after aging at 900 °C [45]. Unfortunately, no explanation was made on how these particles formed. Conventionally, formation of the sigma phase in steels is associated with the presence of a ferrite (bcc) phase and interfaces between the ferrite and austenite (fcc) phases serve as preferred nucleation sites [46]. Alternatively, ferrite can decompose into austenite and sigma [44]. However, the lens-shaped sigma phase particles observed in CoCrFeMnNiV$_{0.5}$ (this work) and in –22Cr–21Ni–6Mo [45] precipitated directly from the fcc phase. The unusual morphology of these particles is likely controlled by specific orientation relationships between the fcc matrix and sigma. Well-known Nenno orientation relationships between the fcc phase and sigma in steels are $(1\,1\,1)\gamma \parallel (0\,0\,1)\sigma$, $[-1\,1\,0]\gamma \parallel [-1\,1\,0]\sigma$ or $(1\,1\,1)\gamma \parallel (0\,0\,1)\sigma$, $[0\,1\,-1]\gamma \parallel [1\,4\,0]\sigma$, which are almost identical (the difference is below 1°) [47]. Other orientation relationships were also reported, namely $(1\,0\,0)\sigma \parallel (1\,0\,0)\gamma$, $[0\,3\,2]\sigma \parallel [0\,1\,1]\gamma$ [48] and $(-1\,1\,1)\gamma \parallel (0\,0\,-1)\sigma$, $[-1\,-1\,0]\gamma \parallel [-1\,-1\,0]\sigma$ or $(-1\,1\,0)\gamma \parallel (-1\,1\,0)\sigma$, $[-1\,-1\,2]\gamma \parallel [1\,1\,3]\sigma$ [45]. However, the obtained (Fig. 6d) orientation relationship, $(2\,2\,0)$fcc $\parallel (-1\,1\,0)\sigma$, $[-2\,-2\,0]$fcc $\parallel [-1\,-1\,3]\sigma$ and $(0\,0\,-2)$fcc $\parallel (3\,3\,2)\sigma$, $[-2\,-2\,0]$fcc $\parallel [-1\,-1\,3]\sigma$, is different from the previously reported. A more detailed analysis of the kinetics of precipitation of the sigma phase from the fcc matrix in the CoCrFeMnNiV$_{0.5}$ alloy is required to understand the mechanism of precipitation that leads to the found crystallographic orientation relationship; however, this is beyond the scope of the current paper.

4.4. Mechanical properties of the CoCrFeMnNiVx alloys

The mechanical properties of the CoCrFeMnNiV$_x$ alloys are significantly affected by the V content only at the V concentrations, at which the hard and brittle sigma phase [49] forms (Fig. 9). One should note that the presence of V atoms in the fcc solid solution of the CoCrFeMnNiV$_{0.25}$ alloy does not result in its hardening according to microhardness measurements as hardness of CoCrFeMnNi and CoCrFeMnNiV$_{0.25}$ alloys in as-solidified state are similar to each other (Table 9). The microhardness values of the CoCrFeMnNiV$_x$ alloys shows complex dependence on the volume fraction of sigma phase (Fig. 9a): microhardness increases with increase of volume fraction of sigma phase, and the increment of hardness becomes higher at higher sigma content. By approximation of the data points in Fig. 9a following power law-type dependence can be obtained:

$$\mathrm{HV} = \upsilon(\mathrm{sigma})^{1.45} + 142.4 \tag{10}$$

where HV is microhardness value of the alloys and υ(sigma) – is volume fraction of sigma phase.

The relationship (10) adequately describes correlation between experimentally measured values of microhardness and volume fraction of sigma phase. According to this relationship, for example, the alloy consisting from sole sigma phase (i.e. υ(sigma) = 100%) would have the microhardness of about 920 HV. This value can be compared to the hardness of the sigma matrix in the CoCrFeMnNiV alloy which equals to 1025–1125 HV. The hardness of sigma phase was measured using smaller load then the overall hardness of the alloys, used in Eq. (9). Generally, the hardness value can be dependent on the applied load [50]. By comparing values of microhardness measurements with different loads in the single

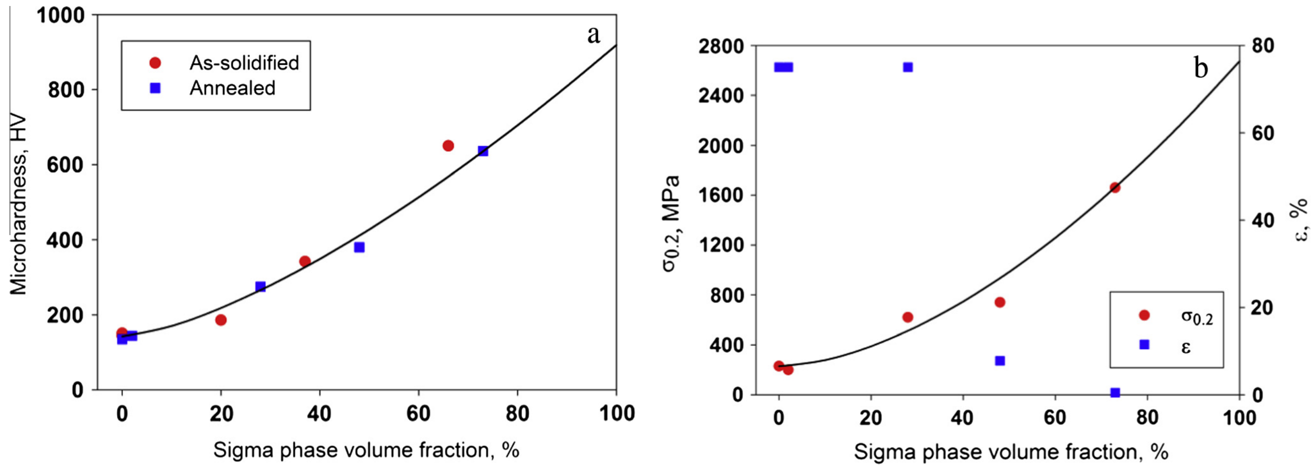

Fig. 9. Dependence of mechanical properties of the $CoCrFeMnNiV_x$ alloys on the volume fraction of sigma phase: (a) microhardness in as-solidified and annealed states; (b) compressive yield strength, $\sigma_{0.2}$, and fracture strain, ε in annealed state.

phase CoCrFeMnNi and $CoCrFeMnNiV_{0.25}$ alloys, it can be deduced that the overall hardness of the alloy is about 0.9 times of the local hardness. Thus the hardness of sigma phase measured with higher load will be about 980 HV, which is reasonably close to the value of 920 HV predicted by Eq. (9). Eq. (10) also correctly predicts contribution from fcc phase, as the predicted hardness od sigma-free alloy (i.e. υ(sigma)) = 0) is 142.4 HV which agrees well with experimentally observed values of 130–170 HV.

Mechanical behavior of the equiatomic CoCrFeMnNi alloy was studied in a number of works [13–15], and our results are in good agreement with those reported previously: the alloy exhibits rather low yield stress value of 230 MPa, pronounced strain hardening stage and large overall ductility. Small addition of V in the $CoCrFeMnNiV_{0.25}$ alloy, has very little effect on the mechanical properties of the alloy; it has similar compressive ductility and slightly lower yield strength of 200 MPa. As the volume fraction of sigma phase in the alloy is insignificant, and sigma phase particles are mostly located on the grain boundaries, it is well anticipated that sigma phase would not result in any substantial strengthening. However, no loss of ductility was observed, possibly both due to insignificant amount of sigma phase and "soft" conditions of compressive testing. Also, the slightly lower yield strength of the $CoCrFeMnNiV_{0.25}$ alloy in comparison with the CoCrFeMnNi alloy indicates that presence of V atoms in fcc solid solution, despite significant distortions caused by V atoms (see Table 10) does not result in any strengthening.

In the case of the $CoCrFeMnNiV_{0.5}$ alloy, which contains 28% of sigma phase, much higher yield strength of 620 MPa is observed. Strengthening due to the large volume fraction of sigma phase is anticipated, especially considering presence of fine lens-shaped particles which should significantly reduce dislocation free paths in the fcc matrix. Nevertheless, the alloy has high ductility of 75%, which is quite surprising because sigma phase is known to be extremely fragile [49]. On the stress–strain curve (Fig. 7) at high strains several drops of stress are observed. These drops might be result of fragile fracture of sigma phase particles. However, it seems that the volume fraction of highly ductile fcc phase is enough for maintaining ductility of the alloy during compressive testing. Unfortunately, ductility significantly reduces in the $CoCrFeMnNiV_{0.75}$ and CoCrFeMnNiV alloys that contain much higher amounts of sigma phase and demonstrate similar brittle behavior. The alloys exhibit high yield strength but very limited strengthening capability.

The dependence of compressive mechanical properties, namely yield strength, $\sigma_{0.2}$, and fracture strain, ε, on the volume fraction of sigma phase is shown in Fig. 9b. The yield strength shows similar to microhardness (Fig. 9a) dependence of power law-type:

$$\sigma_{0.2} = \upsilon(\text{sigma})^{1.69} + 229.3 \tag{11}$$

The flow stress of sigma-free (i.e. υ(sigma) = 0) alloy predicted by Eq. (11) is 229.3 MPa, which is very close to experimentally observed value of 230 MPa. Conventionally, strength (and microhardness) of two-phase alloys is expected to follow the rule of mixtures (ROM), i.e. to be proportional to sum of strength (microhardness) values of individual components in proportion to their volume fractions. In case sigma phase containing $CoCrFeMnNiV_x$ alloys, if simple ROM worked, these dependencies would be close to linear as the contribution from fcc phase is expected to be relatively small. However, dependencies of both strength and microhardness on volume fraction of sigma phase are described by power law-type equations (Eqs. (10) and (11) respectively). As the index of power in both equations is higher than 1 (1.445 in (10) and 1.694 in (11)), it can be suggested that increase of volume fraction of sigma phase cause some additional effect in addition to substituting softer fcc phase. Most probably, this effect is associated with decrease of particle size. It is well accepted that at the same volume fraction, finer particles will promote higher strengthening as the spacing between particles will be smaller. In the studied $CoCrFeMnNiV_x$, increase of V concentration causes not only increase of volume fraction of sigma phase, but also refinement of their size. It causes deviation from ROM observed in Eqs. (10) and (11).

However, these dependencies do not take into account morphology of the particles. Illustration of neglected effect of particle morphology can be found in comparison between the annealed $CoCrFeMnNiV_{0.5}$ and $CoCrFeMnNiV_{0.75}$ alloys. Indeed, values of microhardness and especially yield strength of the $CoCrFeMnNiV_{0.5}$ alloy are reasonably close to those of the $CoCrFeMnNiV_{0.75}$ alloy, despite containing much lower fraction of sigma phase particles. Unpredicted high strength of the annealed $CoCrFeMnNiV_{0.5}$ alloy can be attributed to the presence of lens-shaped particles, which are expected to be effective obstacles for dislocation propagation if the fcc matrix. Therefore, for obtaining optimal combination of mechanical properties in the $CoCrFeMnNiV_x$ alloys, one must not only control volume fraction of sigma phase, but also take into account the morphology of sigma phase particles with lens-shaped morphology being the preferred one.

The dependence of ductility on the volume fraction of sigma phase is non-monotonical. When the volume fraction of sigma phase is below 30%, alloys demonstrate highly ductile behavior and do not fracture even after 75% compression strain. However, one must keep in mind that compression is a soft testing scheme, and, probably, tensile ductility of the alloys containing sigma phase can be different. A rapid decrease in ductility is observed when the volume fraction of sigma becomes ≈50%. Probably, when the volume fraction of fragile sigma phase is beyond certain value, the amount of ductile fcc phase is not sufficient maintain high overall ductility despite presence of brittle sigma phase, like in the CoCrFeMnNiV$_{0.5}$ alloy, and continuous fracture of sigma phase particles results in failure of the whole specimen. When sigma phase becomes the matrix phase, the ductility decreases even further, and is equal to only 0.5%.

5. Conclusions

In current work, microstructure and mechanical properties of the arc melted CoCrFeMnNiV$_x$ ($x = 0$; 0.25; 0.5; 0.75; 1) high entropy alloys were studied in the as-solidified and annealed at 1000 °C for 24 h conditions. Following conclusions were made:

(1) The CoCrFeMnNi alloy is a single-phase fcc solid solution. Alloying with V results in the formation of an intermetallic sigma phase in the alloys with $x \geqslant 0.25$. Sigma phase was found in the CoCrFeMnNiV$_{0.5}$, CoCrFeMnNiV$_{0.75}$ and CoCrFeMnNiV alloys in the as-solidified and annealed conditions. Annealing resulted in an increase in the volume fraction of the sigma phase in alloys with $x = 0.5$, 0.75 and 1.0, and it also appeared in the CoCrFeMnNiV$_{0.25}$ alloy. The volume fraction of the sigma phase increased with V content, from ~2% at $x = 0.25$ to 67–72% at $x = 1.0$.

(2) Sigma phase in the studied CoCrFeNiMnV$_x$ alloys can be predicted using a combination of VEC and δr_V criteria. It is suggested that a combination of VEC in the range of $6.88 \leqslant \text{VEC} \leqslant 7.84$ and $|\delta r_V| > 3.8\%$ is required for the formation of sigma phase.

(3) The sigma phase in the CoCrFeMnNiV$_x$ alloys was found to be significantly enriched with Cr and V. Total atomic concentration of Cr + V in the sigma phase (~39–45% after annealing) was rather insensitive to the composition of the studied alloys. The fcc phase in the sigma containing CoCrFeMnNiV$_x$ alloys also had nearly constant total concentration of Cr + V (~24–25% after annealing). Therefore, the volume fraction of the sigma phase was linearly proportional to the total concentration of Cr + V in the alloys and followed the lever rule between these two critical concentrations. The threshold total concentration of Cr + V required for the formation of the sigma phase in the CoCrFeMnNiV$_x$ alloys is estimated to be 24%.

(4) V was found to be a stronger sigma forming element than Cr in the CoCrFeMnNiV$_x$ alloys and thus less amount of Cr + V was observed in the sigma phase with increasing V content. For example, the Cr + V concentration in the sigma phase decreased from ≈45% in CoCrFeMnNiV$_{0.25}$ to ≈39% in CoCrFeNiMnV after annealing. The Cr equivalent coefficient for V was estimated as 2.55 in the studied CoCrFeMnNiV$_x$ alloys.

(5) Microhadness measurements and compression testing demonstrated that alloying with V above $x = 0.5$ resulted in continuous strengthening (hardening) and loss of ductility of the initially soft and ductile CoCrFeMnNi alloy. For example, microhardness, yield strength and ductility of the CoCrFeMnNi alloys were respectively 135 HV, 230 MPa and more than 75%, and corresponding properties of the CoCrFeMnNiV alloy are 636 HV, 1660 MPa and 0.5%. Power law-type dependencies of yield strength and microhardness on volume fraction of sigma phase were found.

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